CS 4290/CS 6290 *and* ECE 4100/ECE 6100
Advanced/High Performance Computer Architecture
Syllabus - Spring 2021

Prof. Tom Conte conte@gatech.edu

Office Hours: **12:00-1:00pm Wednesdays and Thursdays**
Conte Office Hour Link:
https://gatech.webex.com/gatech/j.php?MTID=m3d9e26290659dcec942fd407b2b87755

Description: 3 Credit Hrs.
Topics concerning the hardware design of computer systems. Advanced techniques in high-performance pipelined central processing units. Memory and I/O systems. Parallel processors including shared-memory multiprocessors and cluster computers. Credit is not allowed for CS 4290, CS 6290, ECE 4100, or ECE 6100.

This course uses Canvas: **http://canvas.gatech.edu** --Check Canvas at least once per week. All assignments course notes will be passed out via Canvas only.

Preq: **(1)** Understanding of basic computer organization, equivalent of GT CS 2200
(2) Competence with programming (including data structures)
(3) Programming ability in C or C++ (**only**); and, *some* knowledge of an assembly language

Book: ***No BOOK*** **-** Course notes only. At times I may also assign papers to read.

Goals: By the conclusion of this course, you will appreciate major topics in the field of Computer Architecture. You will be able to read and understand current research papers in the field.

Remarks: This is my research area. ***This course is very challenging*** because the topics are highly technical. I have a reputation for being a very fair professor. Note I will be traveling a lot during this semester. Instead of canceling lectures, I will ask a senior graduate student to give those lectures.

Topics [notes volume name]:

1. [n1_intro] Introduction, overview and history of general-purpose computers, current trends
2. [n2_perf] Measuring performance and cost
3. [n3_cache] Uniprocessor memory hierarchy design
4. [n4_pipe] Pipelined microarchitectures
5. [n5_ilp] Instruction level parallel microarchitectures: superscalar and VLIW
6. [n6_multiproc] Multiprocessors: shared memory vs. message passing, Cache-coherent shared memory: snooping and directory-based protocols, Memory consistency
7. [n7_sync] Synchronization
8. [n8_interconn] Interconnection networks and storage
9. Advanced topics (time permitting): reliability, VLSI overview, GPUs, I/O, DRAM/PCM

Course Workload and Grade Calculation:
The course has three programming/simulation projects.

50%	Projects I (5%) II-IV (15% each)
10%	Exam I (**Thursday, February 18th, online**)
10%	Exam II **(Thursday, April 1th, online)**
20%	Final exam (**Thursday, May 6, 8:00 AM - 10:50 AM, online)**
10%	Homeworks (approximately 4 to 5)

Attendance at all exams is MANDATORY. Drop the class now if you cannot attend all three exams.

CS 4290/ECE 4100 final grades will use the scale:
[100:90] = A, (90,80] = B, (80,70] = C, (70,60] = D, (60,0] = F.
CS 6290/ECE 6100 final grades will use the scale:
[100:85] = A, (85,70] = B, (70,60] = C, (60,50] = D, (50,0] = F.

Class Policies

1. **Lectures: Class Attendance is Very Important!** I do not use powerpoint as I find it ineffective. Instead, I teach by writing 'on the board.' I encourage interactive lectures. Feel free to ask questions if you do not understand something. My goal is to make the lectures ones you truly want to attend.
2. **No distractions during lecture:** ***avoid surfing, reading news, social media, etc.*** Use a paper notebook to take notes. Research has proven this approach enhances learning and retention.
3. **There will be no extra credit given in this class.**
4. **Projects:** There are four semester projects that involve programming a simple simulator of a computer architecture component. More details of the projects will be announced when each is assigned. **Project IV will be due during Final Instructional Class Days.**
5. **Exams:** There are 2 semester exams. These will be several short questions, closed book, closed notes. Exams are during lecture time. We will use Honorlock in this course.
6. **Homework:** Generally, homework assignments are difficult but not impossible. They cover material from lecture. When they are assigned concurrent with active projects, they also will emphasize the concepts needed for the projects. There will be approximately four to five homework assignments in total. The final homework will be due during Final Instructional Class Days.
7. **Homework #0:** In an effort to get to know you and to make sure you can find my office: you are required to visit my office at least once during my office hours and before the second exam (April 2nd). You must announce that your visit is "for homework 0." This counts as a real homework! *If for some reason you cannot meet this requirement, please contact me via email.*
8. **Posting of Notes:** Generally class notes are posted online after a section of the course outline has been completed. For longer sections (e.g., n6_multiproc), I'll post partial notes before the end of the section.

9. **Statement on late assignments:** ***No work will be accepted late***. All assignments must be turned in by the due date and time. Students who believe they have valid excuses to miss exams must comply with Institute excused absence regulations: http://studentlife.gatech.edu/content/class-attendance

10. **Regrading**: We only re-grade an assignment if you answered something **correctly** but it was marked as **incorrect**. ***There are no re-grades for higher partial credit.*** Re-grade requests must be submitted within three (3) calendar days of the return of the graded assignment or exam.

11. **Statement for students with disabilities:** Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Office of Disability Services: http://disabilityservices.gatech.edu

12. **Statement for not mentioned policies and issues:** Any policies and issues not mentioned in this syllabus will follow policies and procedures of the Georgia Institute of Technology: http://policylibrary.gatech.edu

Coding guidelines (C/C++ code)

13. You must turn in ALL files specified in the "Deliverables" section of the assignment instructions. We reserve the right to impose a penalty on submissions that do not follow the given submission directions.
14. You must provide a ***Makefile*** that compiles and links your code by default. **If you are given a Makefile with the project, we expect your code to compile under the given Makefile.**
15. Your code must compile with **gcc** on Ubuntu 18.04 LTS or later. We will provide a virtual machine to use. If your code does not compile, you will receive a 0 for the assignment.
16. You will be penalized if your code produces warnings when compiled with the given Makefile, or the following flags if no Makefile is provided: **gcc -Wall -pedantic -O2**
17. Code should be well commented and use a clean, consistent (readable) style (i.e., proper indenting, etc.). We reserve the right to impose style requirements, and deduct for non-conforming solutions. This is **not** the obfuscated C code competition!

Unauthorized Copying and Plagiarism (Cheating)

18. As a Georgia Tech student, are assumed to have read and agreed to the Georgia Tech Honor Code, see http://osi.gatech.edu/content/honor-code
19. A student must submit an assignment or project as his/her own work (this is what is expected of the students).
20. **Important:** collaboration is not permitted. **No code or answers should be copied from others. Such copying is unauthorized and will be referred to the Office of Student Integrity. Simply put, the sharing of the answer is academic misconduct**. If you are not sure about it, please ask a TA or stop by the instructor's office during the office hours.
21. Suspected plagiarism will be reported to the Division of Student Life Office of Student Integrity. It will be prosecuted to the full extent of Institute policies.

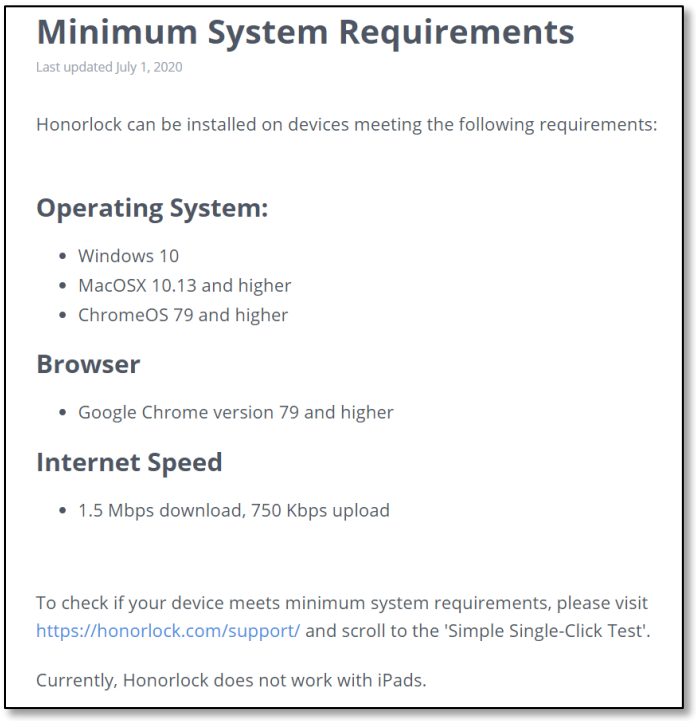
22. **Using code from GitHub, via Googling, from Stack Overflow, etc., is plagiarism and is not permitte**d. Also, **do not publish your assignments on public repositories or chats (i.e., accessible to other students)**. This is also a punishable offense treated as plaigerism.
23. TAs and Instructor determine whether the project is plagiarized. **Trust us, it is *really easy* to determine this**….

Statement for not mentioned policies and issues: Any policies and issues not mentioned in this syllabus will follow policies and procedures according to the Georgia Institute of Technology: http://policylibrary.gatech.edu

HonorLock

This course will use digital proctoring for all quizzes and exams. The following are required of students.

Honorlock technical requirements:



Minimum System Requirements

Last updated July 1, 2020

Honorlock can be installed on devices meeting the following requirements:

Operating System:

- Windows 10
- MacOSX 10.13 and higher
- ChromeOS 79 and higher

Browser

- Google Chrome version 79 and higher

Internet Speed

- 1.5 Mbps download, 750 Kbps upload

To check if your device meets minimum system requirements, please visit https://honorlock.com/support/ and scroll to the 'Simple Single-Click Test'.

Currently, Honorlock does not work with iPads.

- Students must have a **broadband internet connection** (check at https://honorlock.com/support under "Simple Single-Click Test")
- Students must have a **webcam and microphone**
- Students must have a **secure private location to take an exam**
- Students will be asked to **provide a picture ID** and **take a picture of themselves via a webcam** as part of the exam process
- Honorlock ***is not*** compatible with Linux OS, Virtual Machines, tablets, or smartphones
- Honorlock **requires** the installation of Google Chrome and the Honorlock Chrome extension

COVID-19 Expectations and Guidelines

Each of us has a responsibility to ourselves and our fellow Yellow Jackets to be mindful of our shared commitment.

Instructor Illness or Exposure to Covid-19

During the spring 2021 semester, some faculty members may be required to quarantine due to exposure or isolate due to a Covid-19 diagnosis. Some disruption to classes or services is inevitable, but Georgia Tech is making every effort to ensure continuity of operations. As is the case in any semester, faculty may cancel a class if they have an illness or emergency situation and cover any missed material at their own discretion. If an instructor needs to cancel a class, they should notify students as early as possible.

Student Illness or Exposure to Covid-19

During the semester, you may be required to quarantine or self-isolate to avoid the risk of infection to others. Quarantine is the separation of those who have been exposed to someone with Covid-19 but who are not ill; isolation is the separation of those who have tested positive for Covid-19 or been diagnosed with Covid-19 by symptoms.

If you have not tested positive but are ill or have been exposed to someone who is ill, please follow the Covid-19 Exposure Decision Tree for reporting your illness.

During the quarantine or isolation period you may feel completely well, ill but able to work as usual, or too ill to work until you recover.

Unless you are too ill to work, you should be able to complete your remote work while in quarantine or isolation.

If you are ill and unable to do course work this will be treated similarly to any student illness. The Dean of Students will have been contacted when you report your positive test or are told that it is necessary to quarantine and will notify your instructor that you may be unable to attend class events or finish your work as the result of a health issue. Your instructor will not be told the reason.

CARE Center, Counseling Center, Stamps Health Services, and the Student Center

These uncertain times can be difficult, and many students may need help in dealing with stress and mental health. The **CARE Center** and the **Counseling Center**, and **Stamps Health Services** will

offer both in-person and virtual appointments. Face-to-face appointments will require wearing a face covering and social distancing, with exceptions for medical examinations. Student Center services and operations are available on the **Student Center** website. For more information on these and other student services, contact the Vice President and Dean of Students or the **Division of Student Life**.

Accommodations for Students at Higher Risk for Severe Illness with Covid-19

Students may request an accommodation through the Office of Disability Services (ODS) due to 1) presence of a condition as defined by the Americans with Disabilities Act (ADA), or 2) identification as an individual of higher risk for Covid-19, as defined by the Centers for Disease Control (CDC). Registering with ODS is a 3-step process that includes completing an application, uploading documentation related to the accommodation request, and scheduling an appointment for an "intake meeting" (either in person or via phone or video conference) with a disability coordinator.

If you have been approved by ODS for an accommodation, I will work closely with you to understand your needs and make a good faith effort to investigate whether or not requested accommodations are possible for this course. If the accommodation request results in a fundamental alteration of the stated learning outcome of this course, ODS, academic advisors, and the school offering the course will work with you to find a suitable alternative that as far as possible preserves your progress toward graduation.